Seismological Society of America

SSA 2026 Workshop

Introduction to DAS for Seismology: From Data Acquisition to Analysis

DAS Basics: Instruments & Measurements

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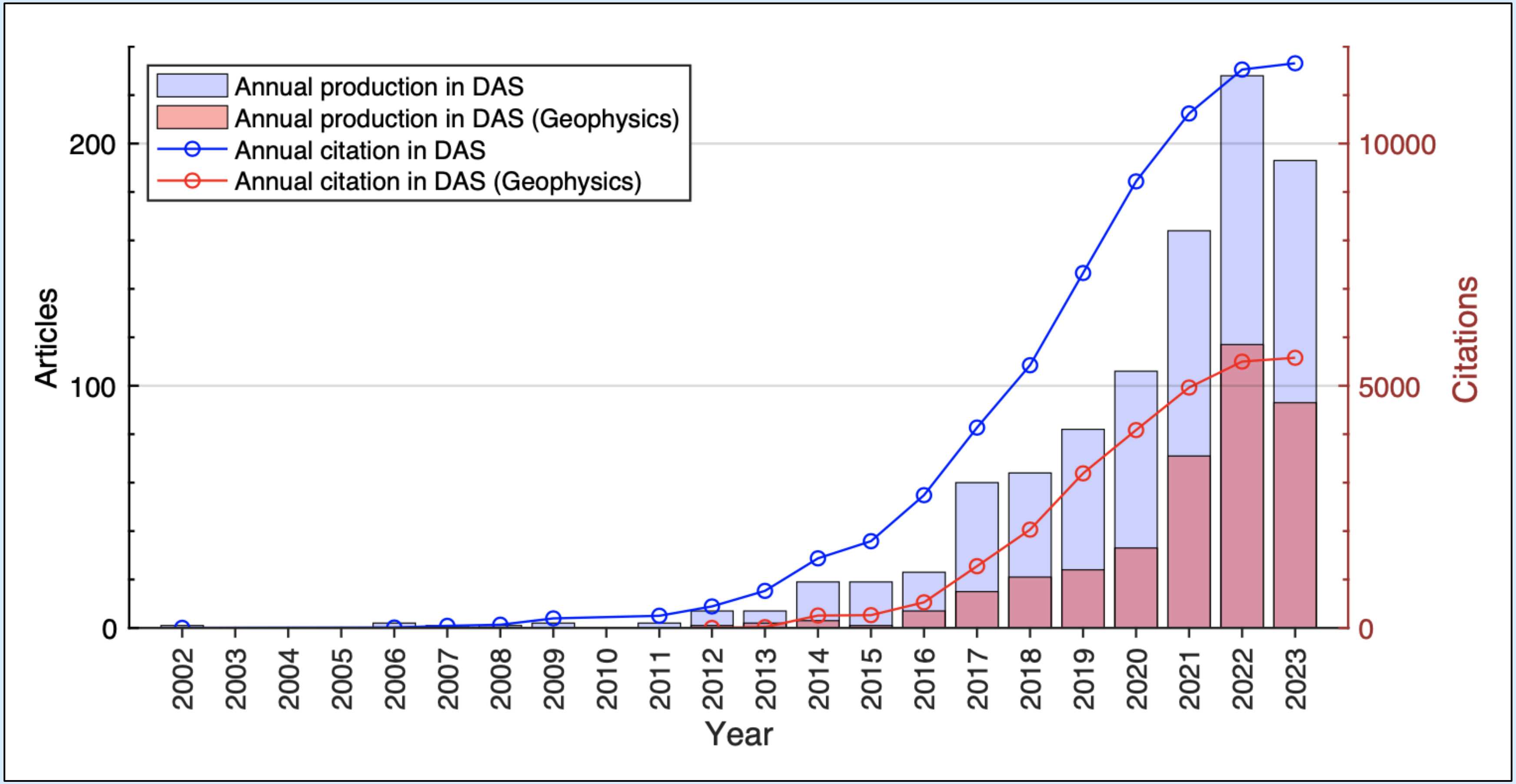
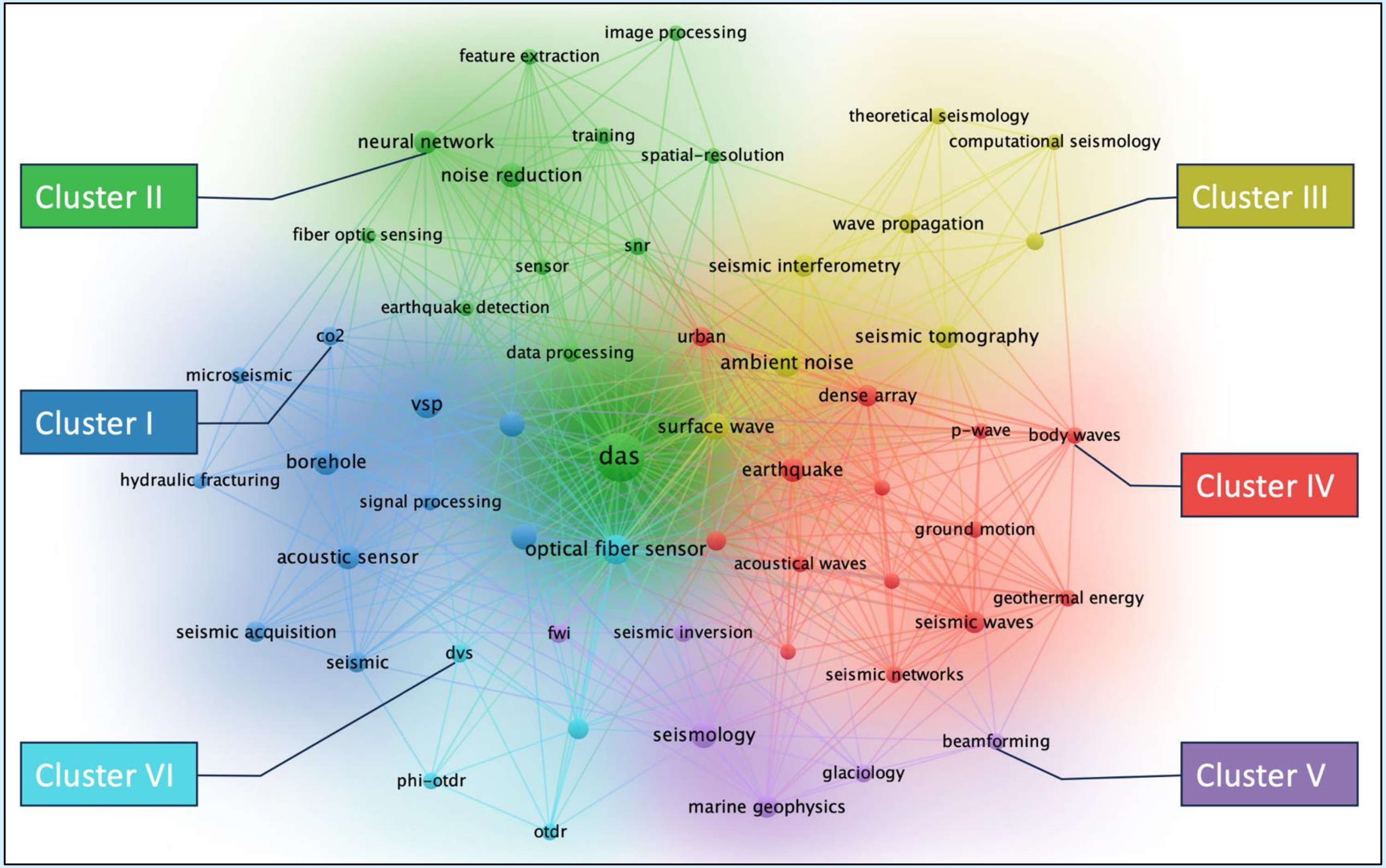
(alphabetical order)

April 14th, 2026

The scientific impact of DAS

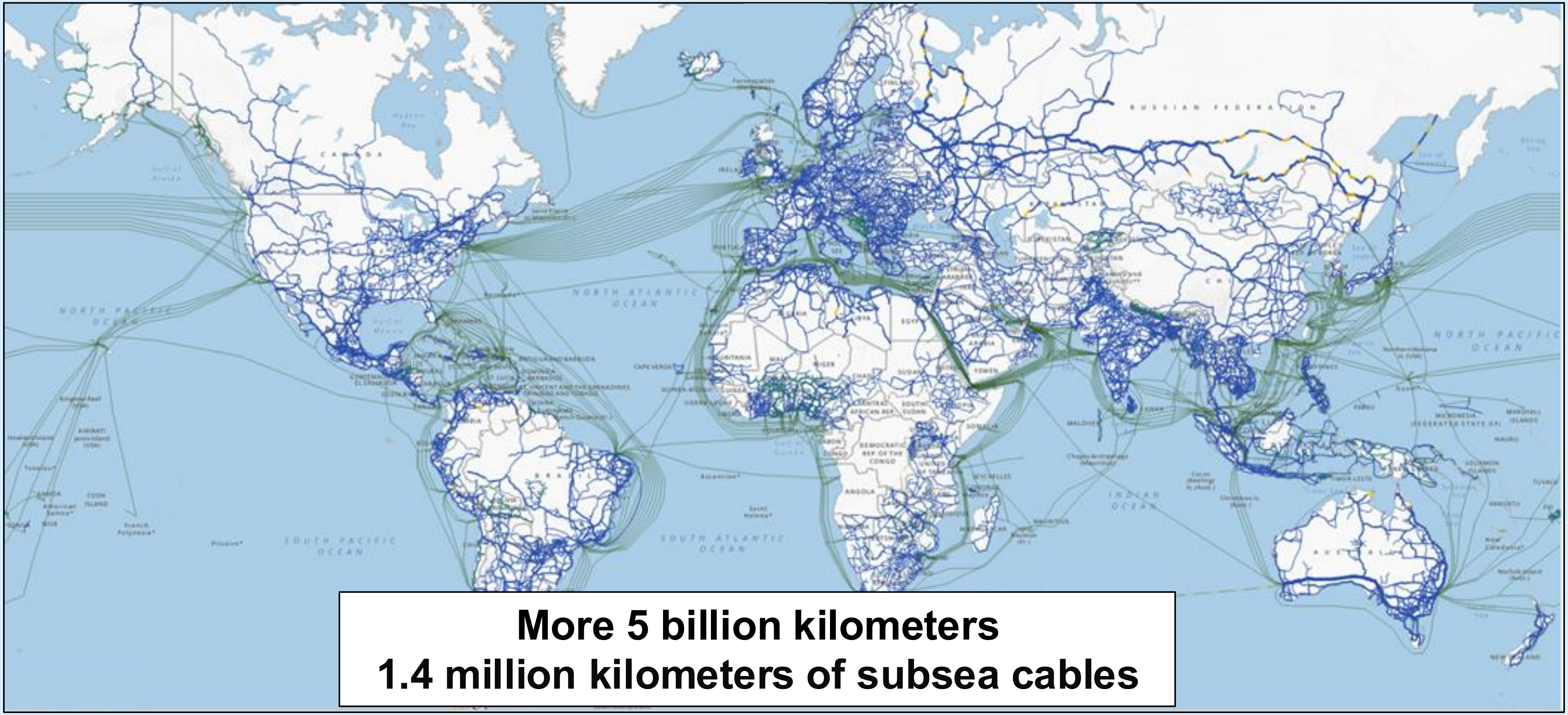


The scientific impact of DAS



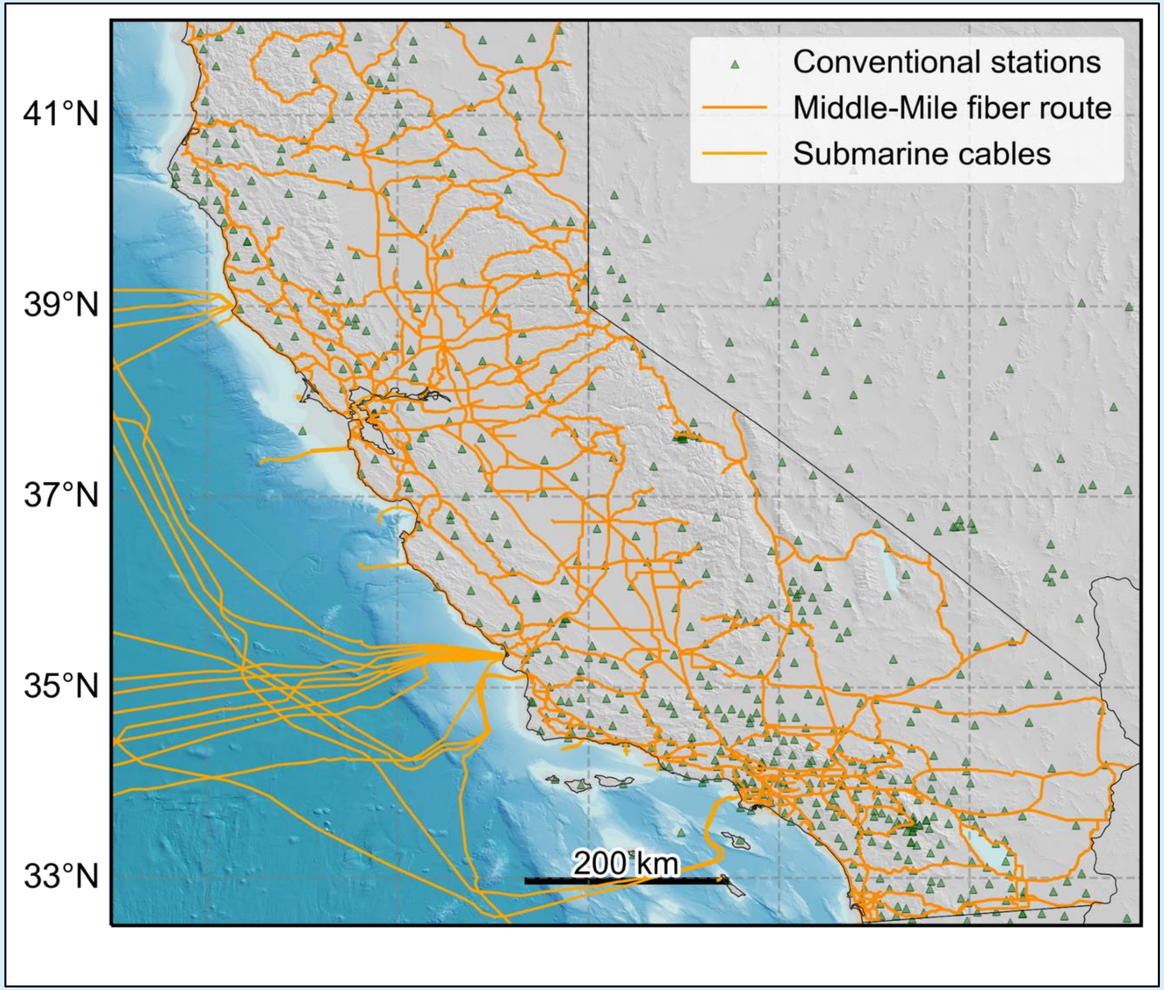
Cheng (2024)

A pervasive infrastructure



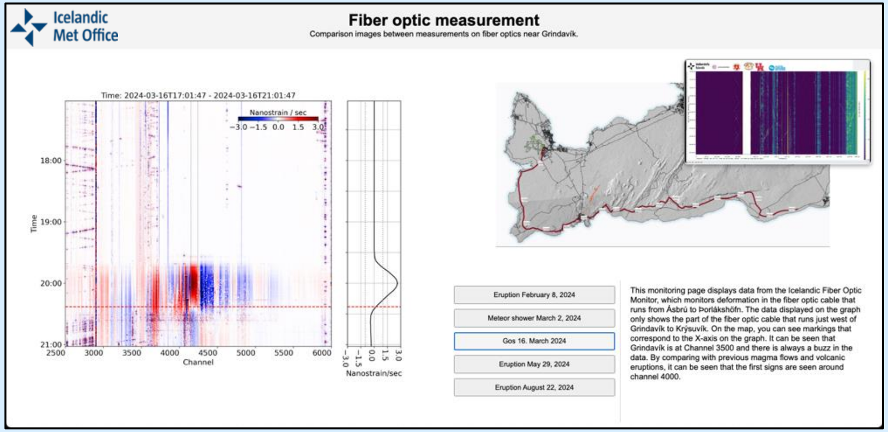
More 5 billion kilometers
1.4 million kilometers of subsea cables

A pervasive infrastructure

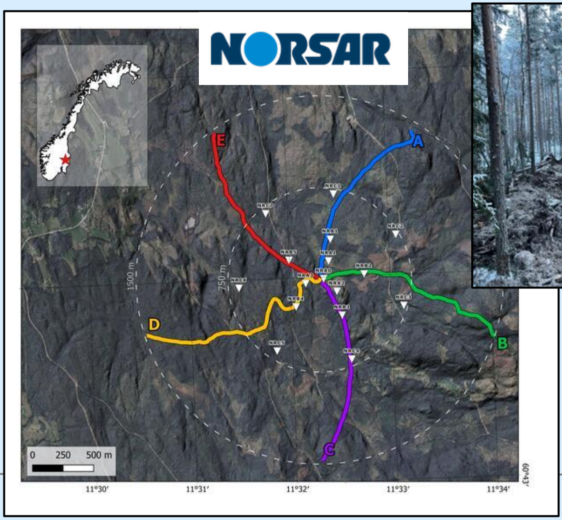


DAS in monitoring operations

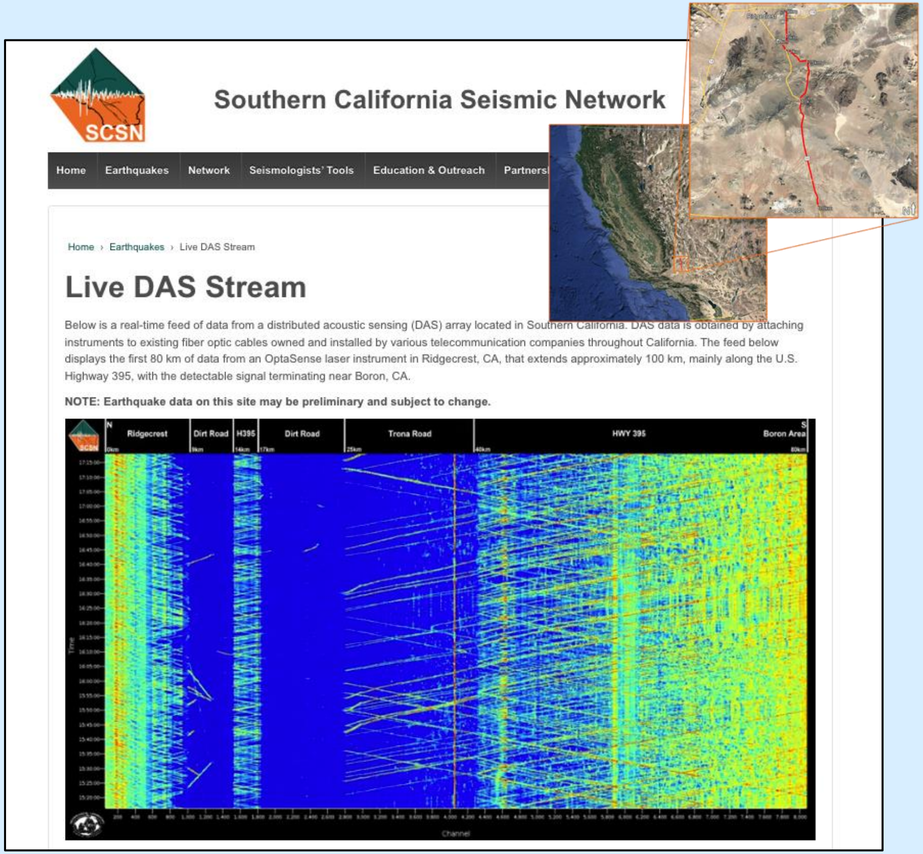
Eruption early warning

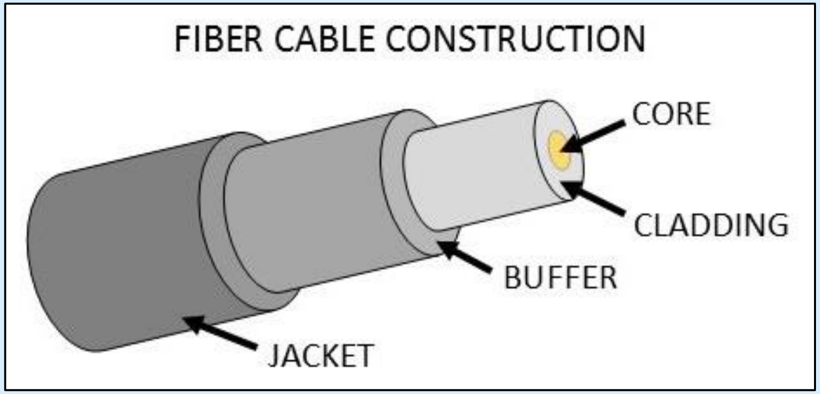


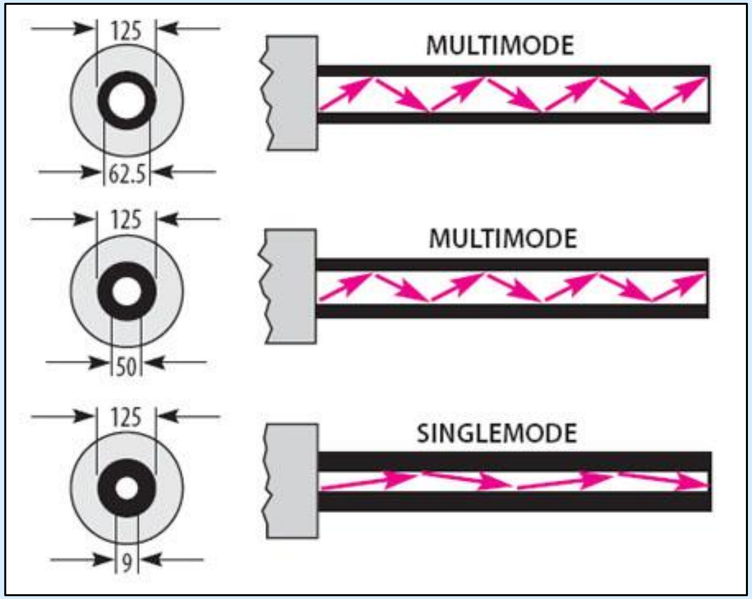
Explosion detection

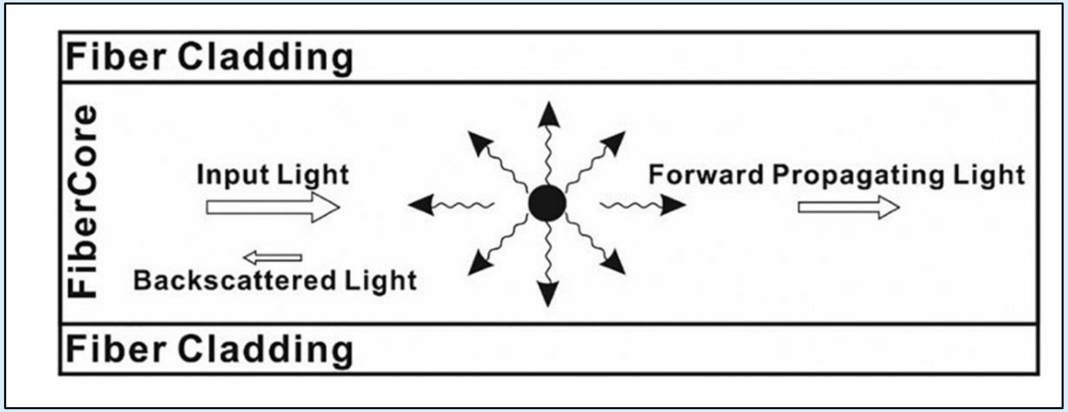


Earthquake monitoring



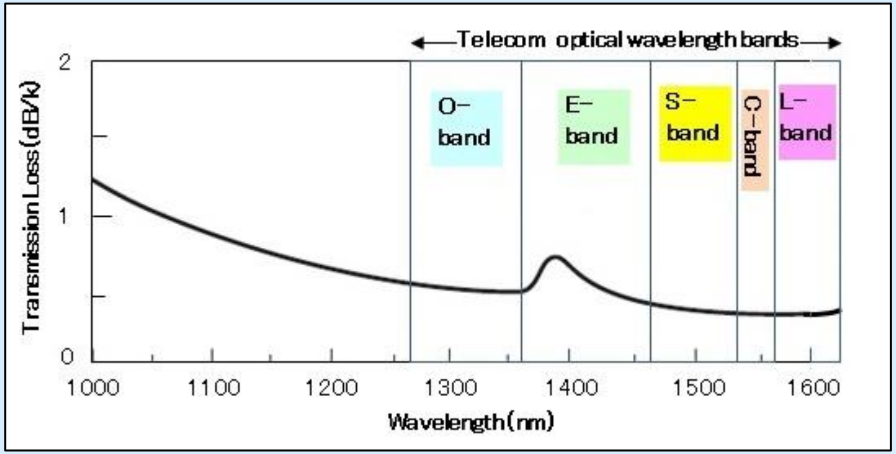

FIBER CABLE CONSTRUCTION
CORE
CLADDING
BUFFER
JACKET

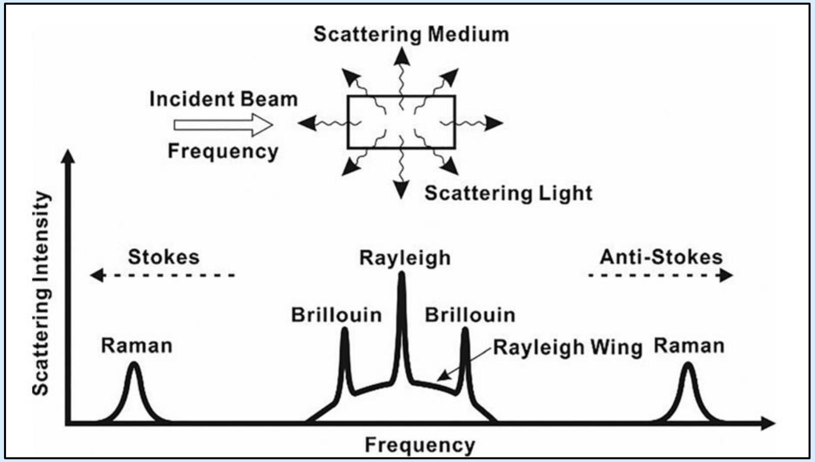

125
62.5
MULTIMODE
125
50
MULTIMODE
125
9
SINGLEMODE


Fiber Cladding
FiberCore
Input Light
Forward Propagating Light
Backscattered Light
Fiber Cladding

Lu et al. (2019)

Laser scattering mechanisms





Lu et al. (2019)

Fiber-sensing: OTDR

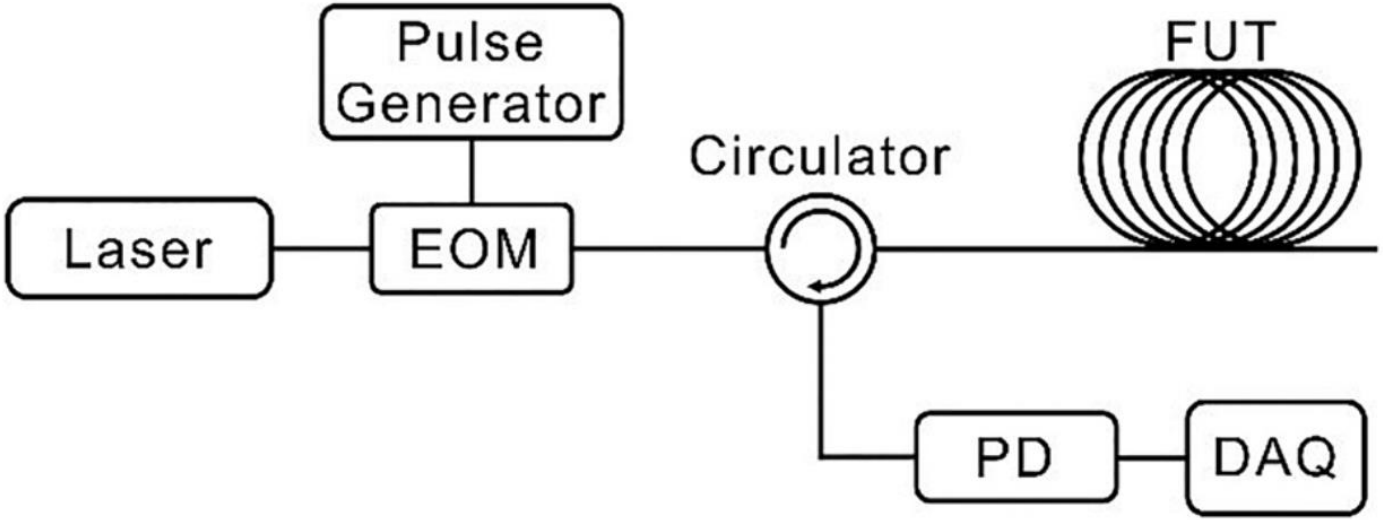
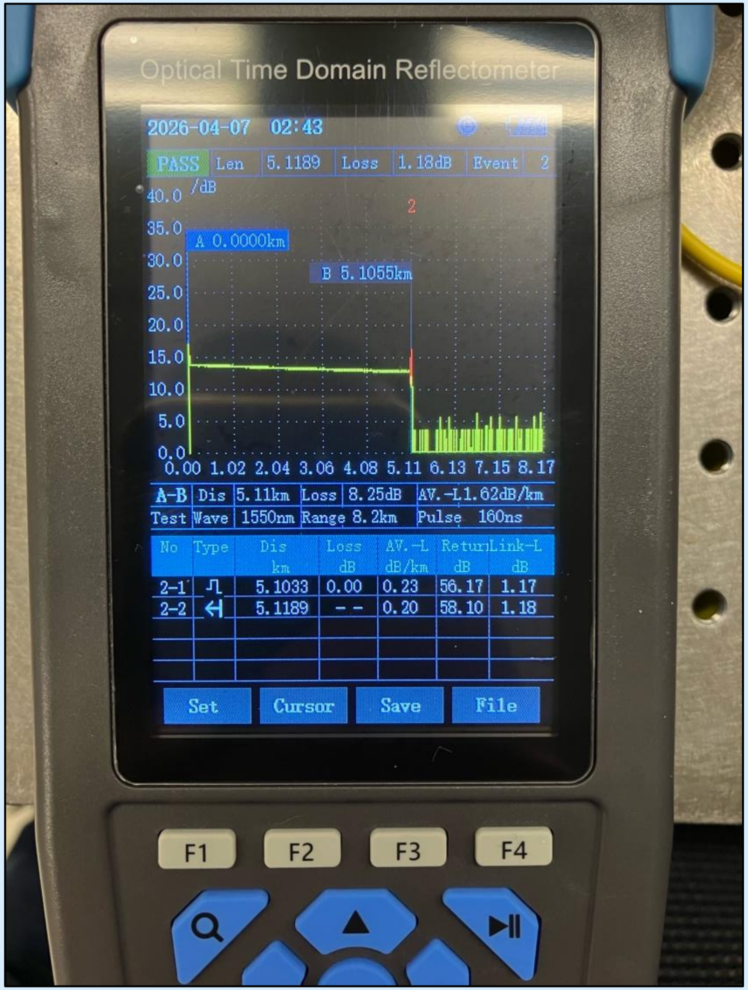


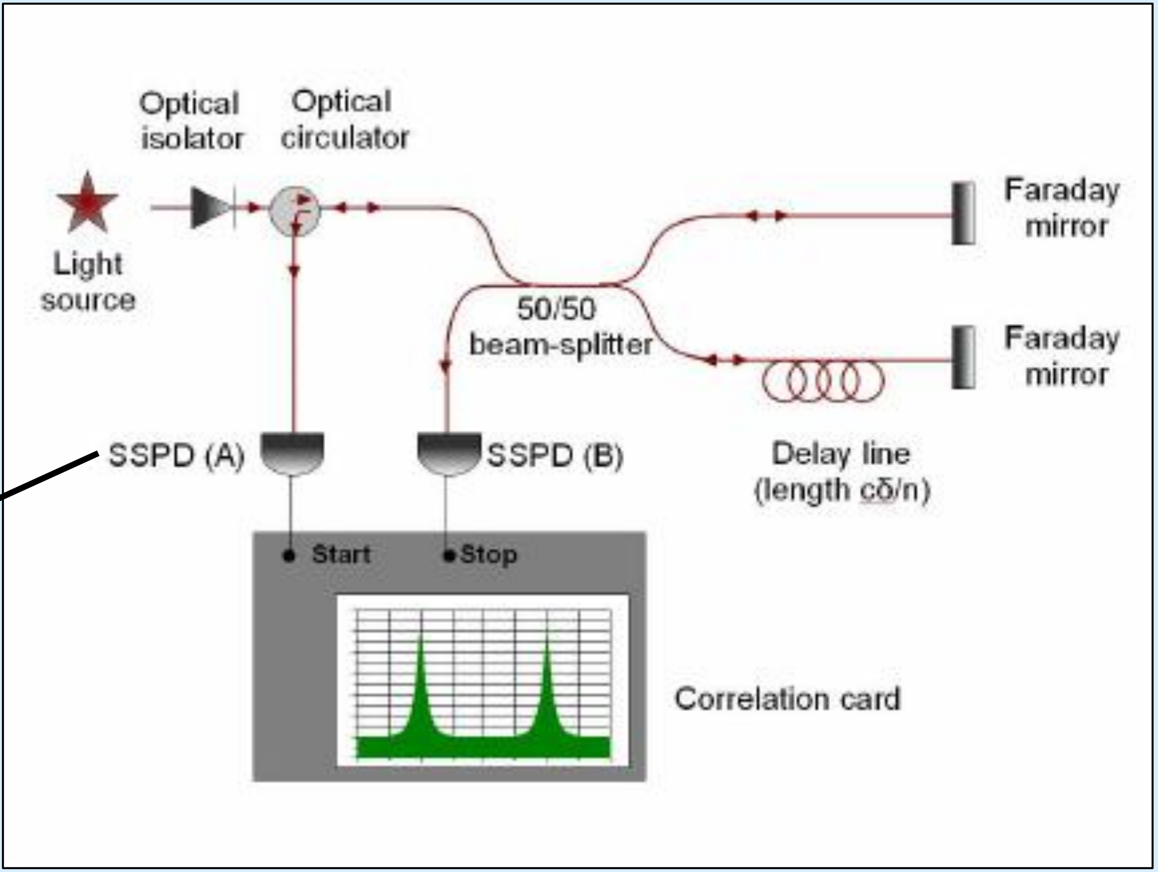
FIG. 3. A standard OTDR configuration. EOM: electro-optic modulator; PD: photodiode; DAQ: data acquisition device; FUT: fiber under test.

Lu et al. (2019)

Fiber-sensing: OTDR



Fiber-sensing: Unbalanced Michelson interferometer



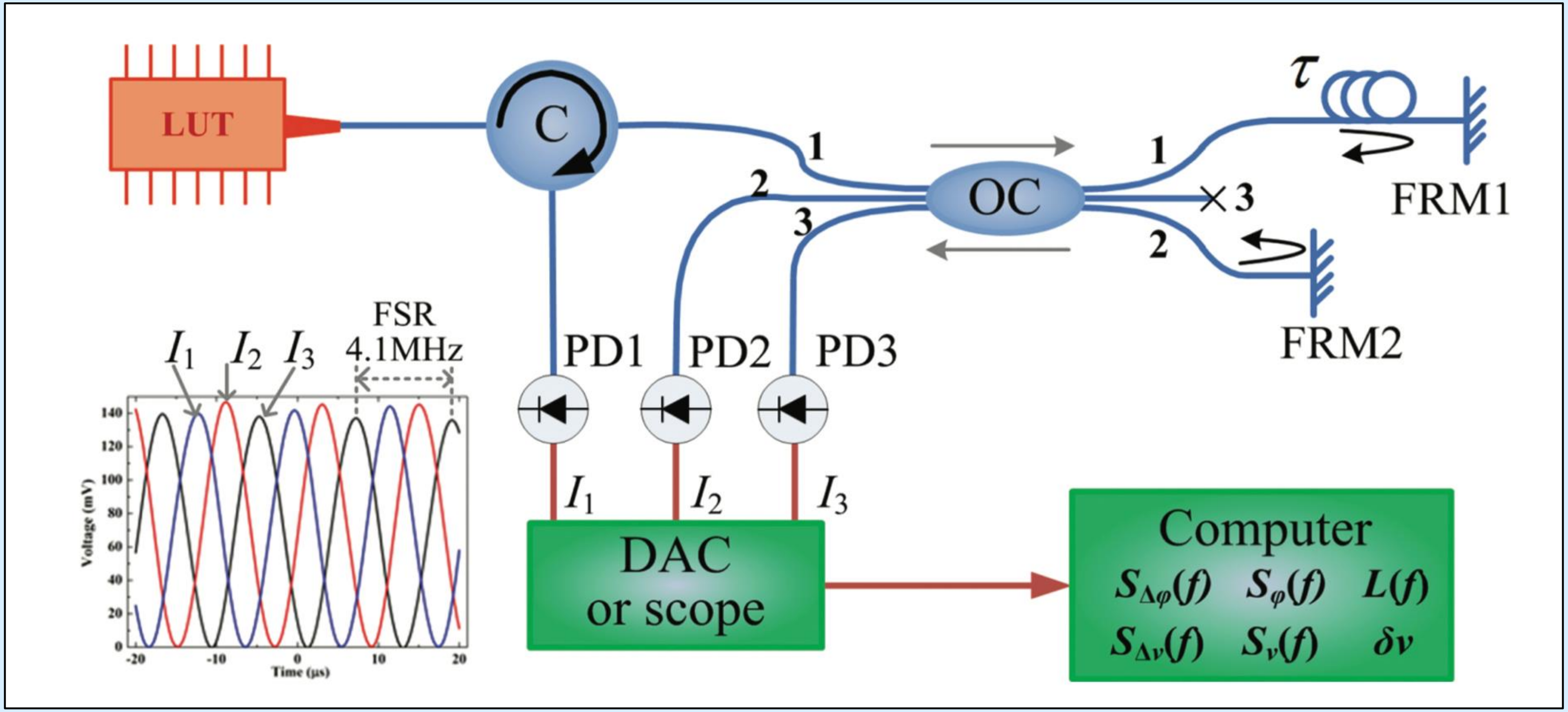
$$E_1 = \sqrt{P_1} e^{-i(\omega t - \phi_1)}$$

$$E_2 = \sqrt{P_2} e^{-i(\omega t - \phi_2)}$$

$$I_1 = |E_1 + E_2| = \frac{1}{2}\{P_1 + P_2 + \boxed{2\sqrt{P_1 P_2}\cos[\Delta\phi(t)]\}}$$

Lebreton et al. (2013)

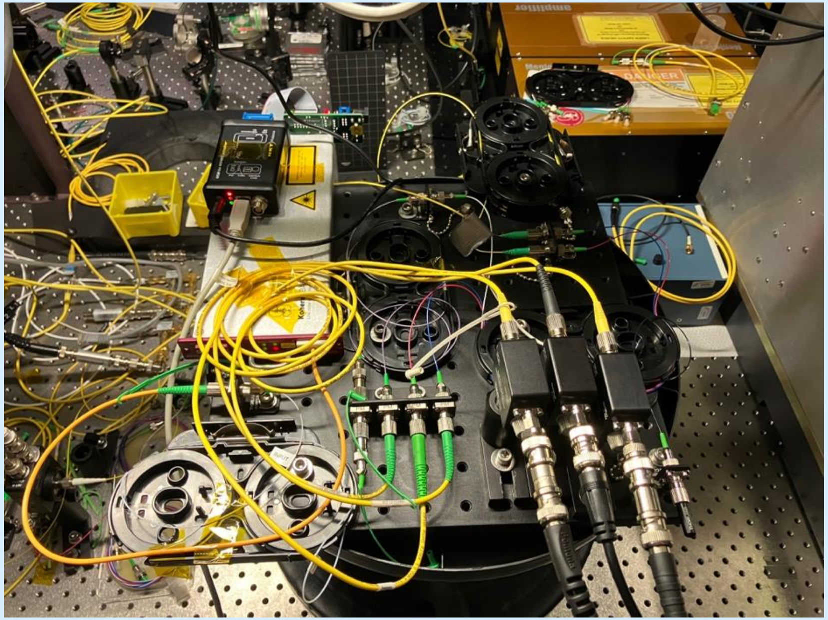
Optical schematic

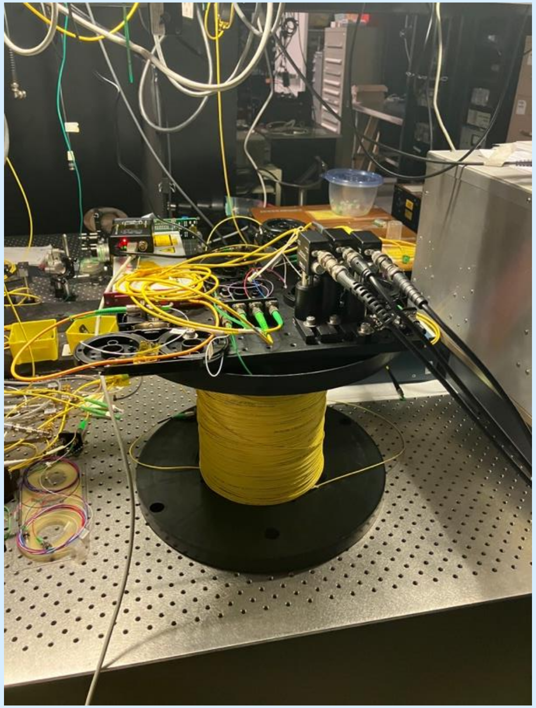


Fei et al. (2016)

Fiber-sensing: Unbalanced Michelson interferometer

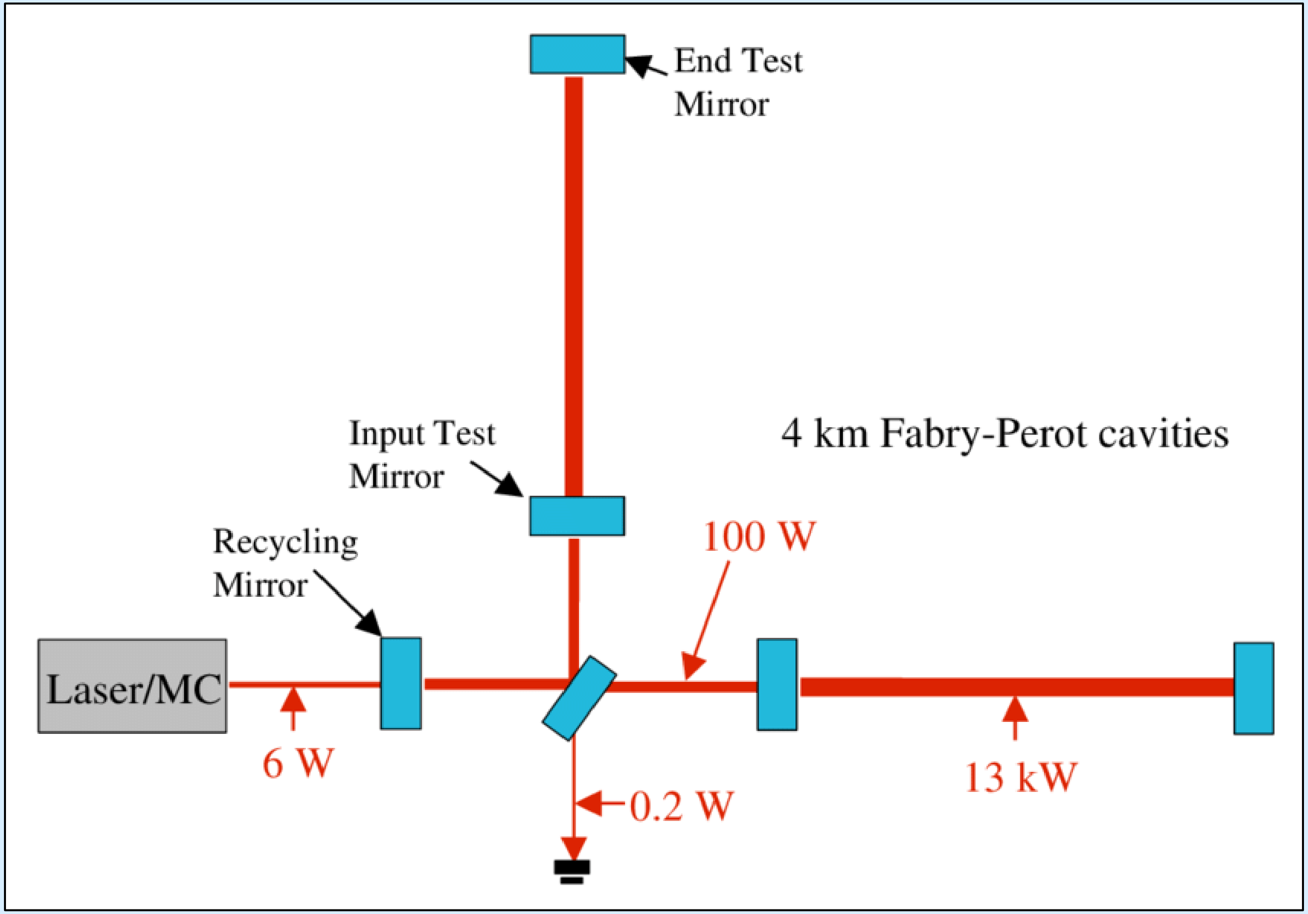
$$I_1 = |E_1 + E_2| = \frac{1}{2}\{P_1 + P_2 + +2\sqrt{P_1 P_2}\cos[\Delta\phi(t)]\}$$





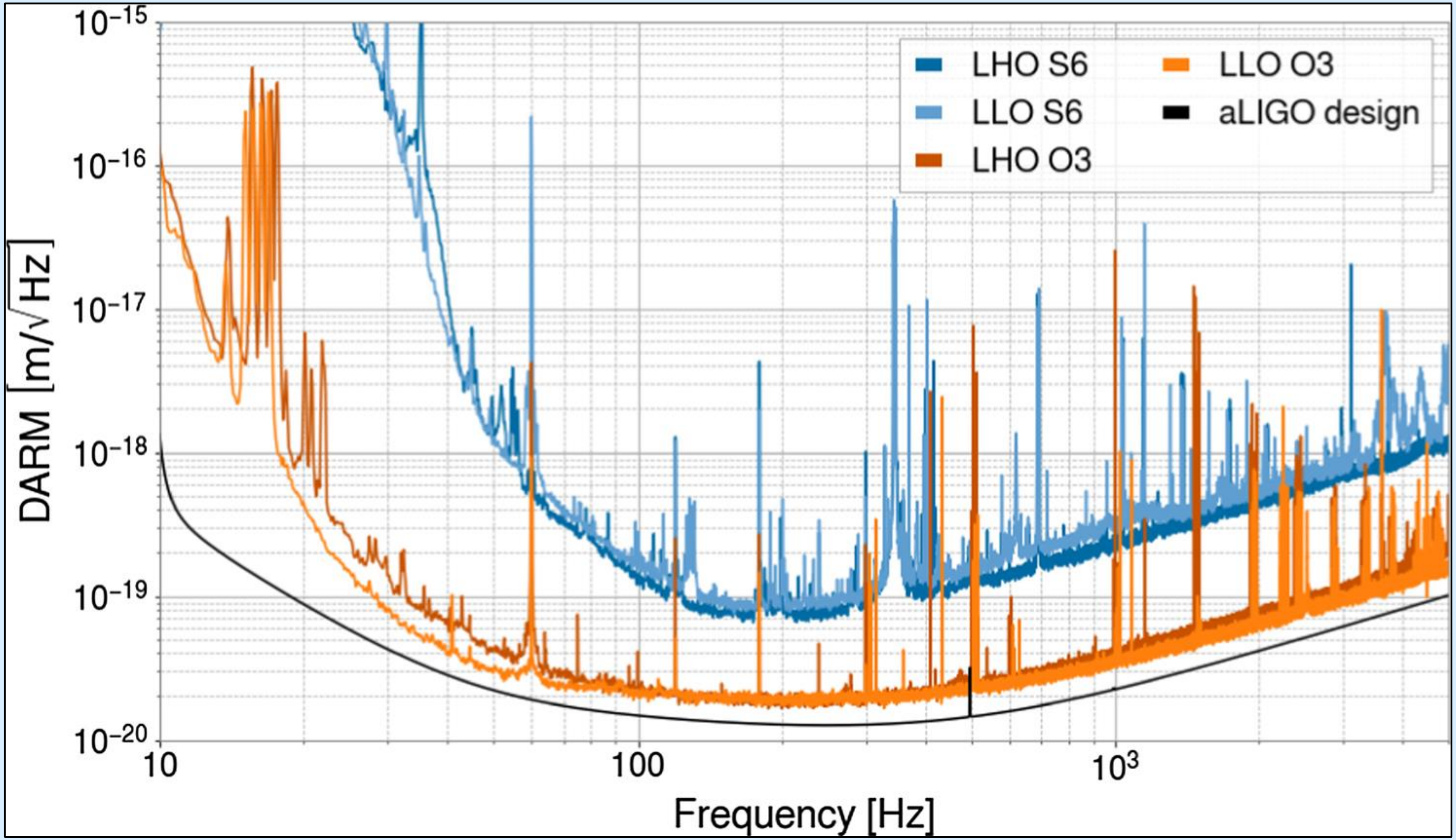
Collaboration with Leo Hollberg Lab

Laser interferometry: LIGO



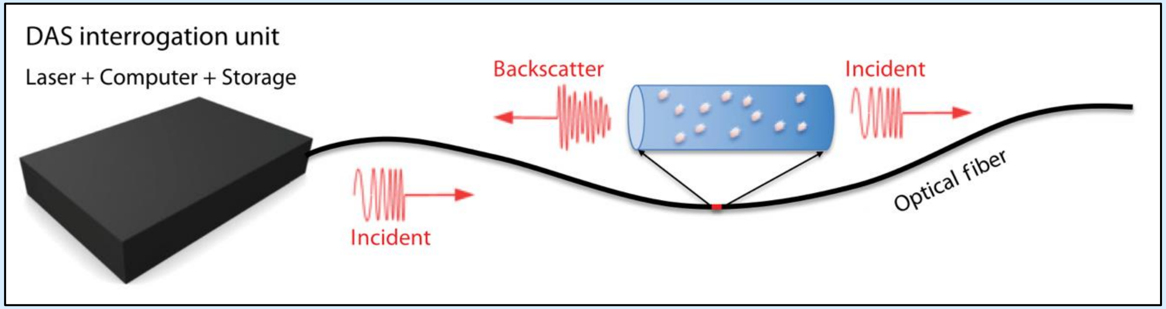
Hamilton et al. (2005)

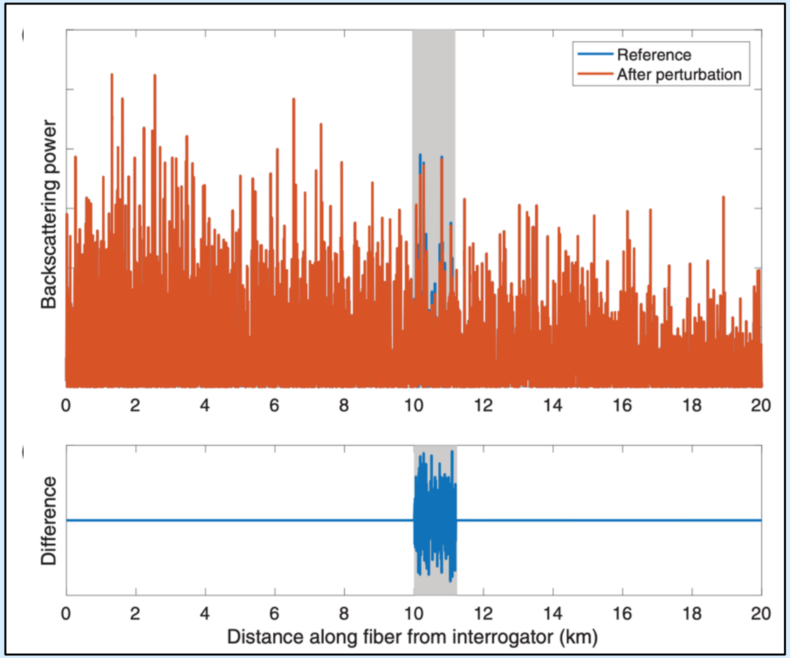
Laser interferometry: LIGO



Nguyen et al. (2016)

Fiber-sensing: DAS





Zhan, 2020

Fiber-sensing: DAS

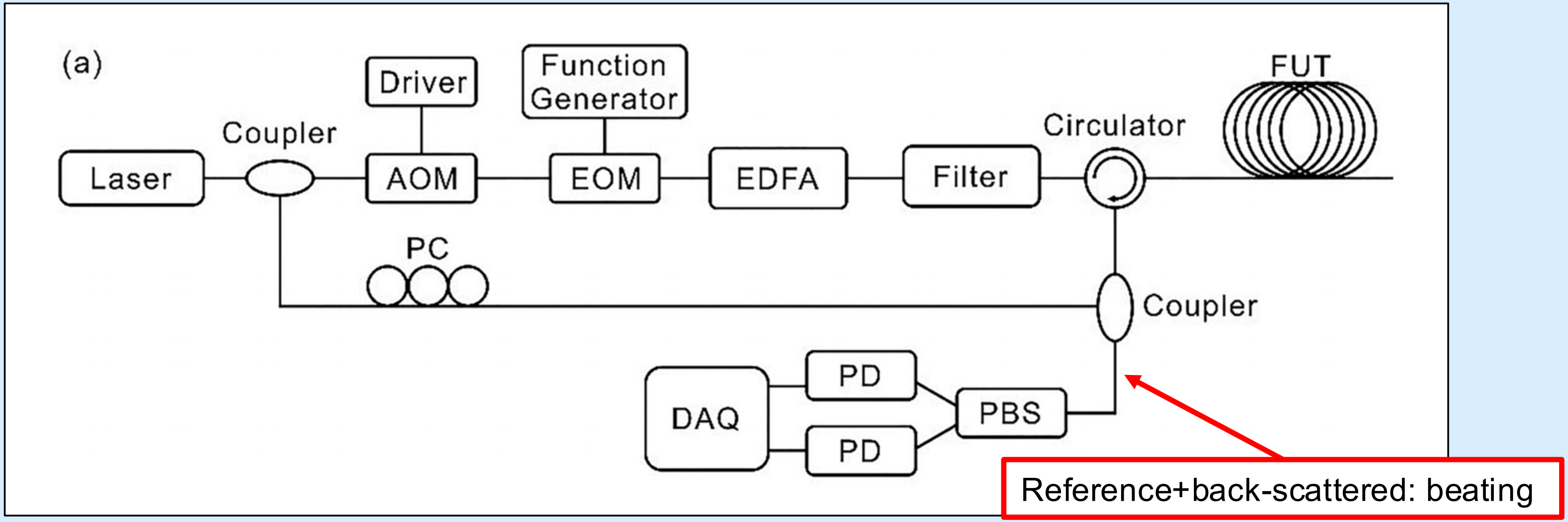
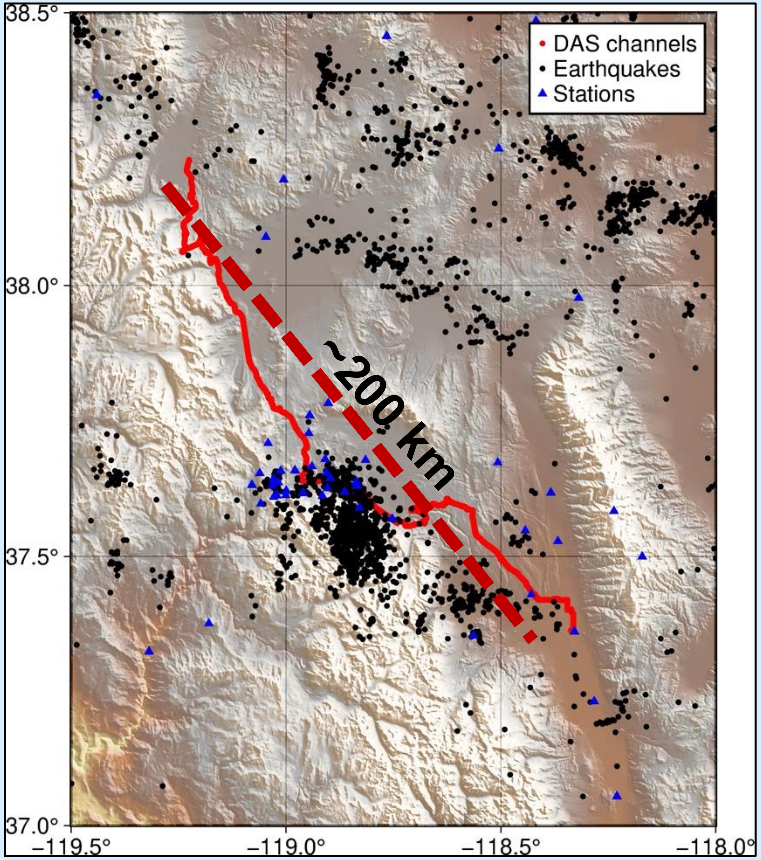
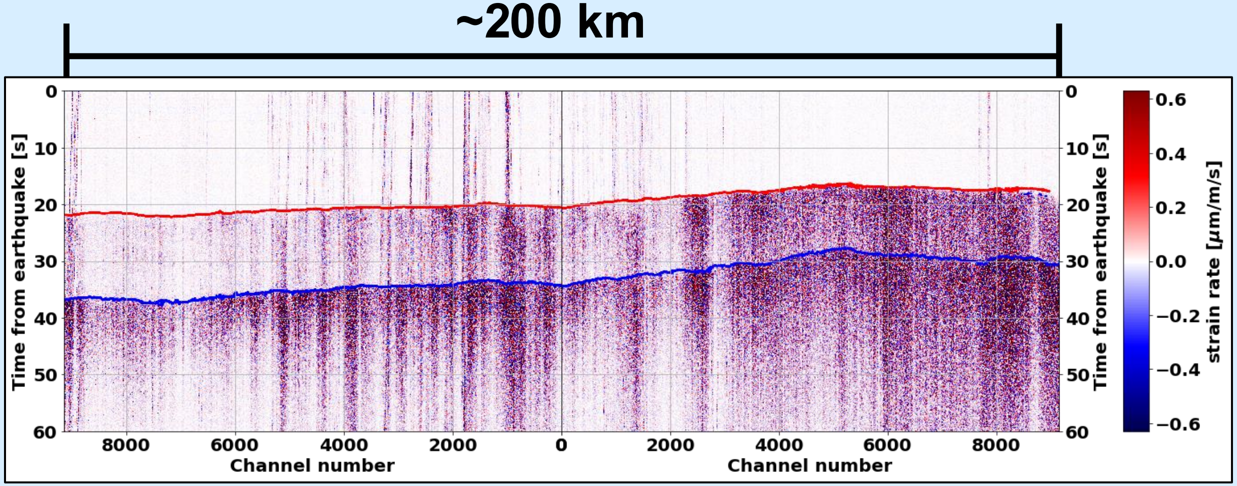


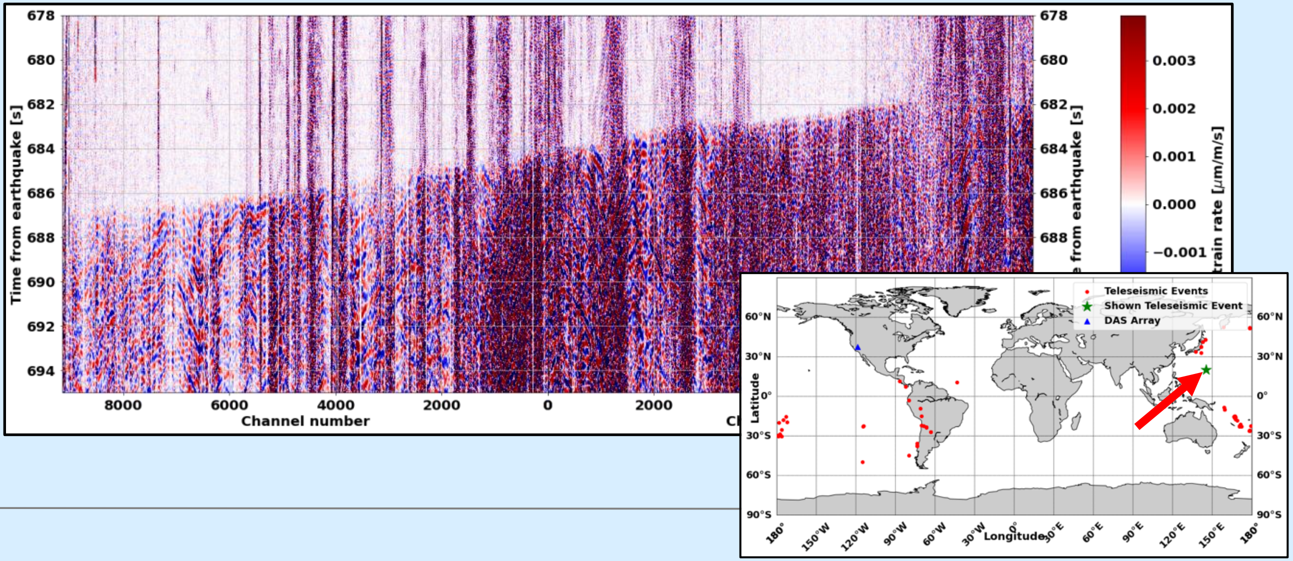
FIG. 4. Experimental setups of the Φ-OTDR system for vibration amplitude measurement. (a) Coherent detection and (b) direct detection. AOM: acousto-optic modulator; EDFA: erbium-doped fiber amplifier; PC: polarization controller; PD: photodiode; PBS: polarization beam splitter; DAQ: data acquisition device; FUT: fiber under test.

Lu et al. (2019)

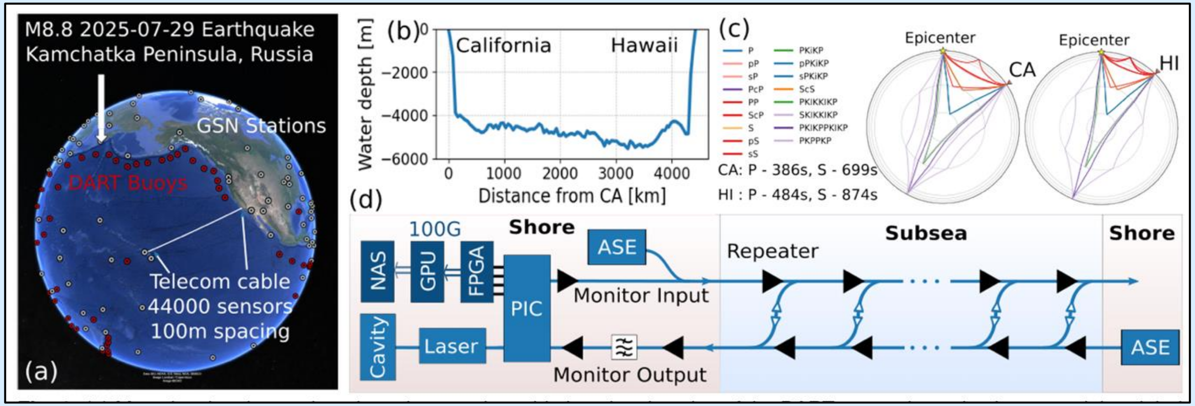
Long-range arrays

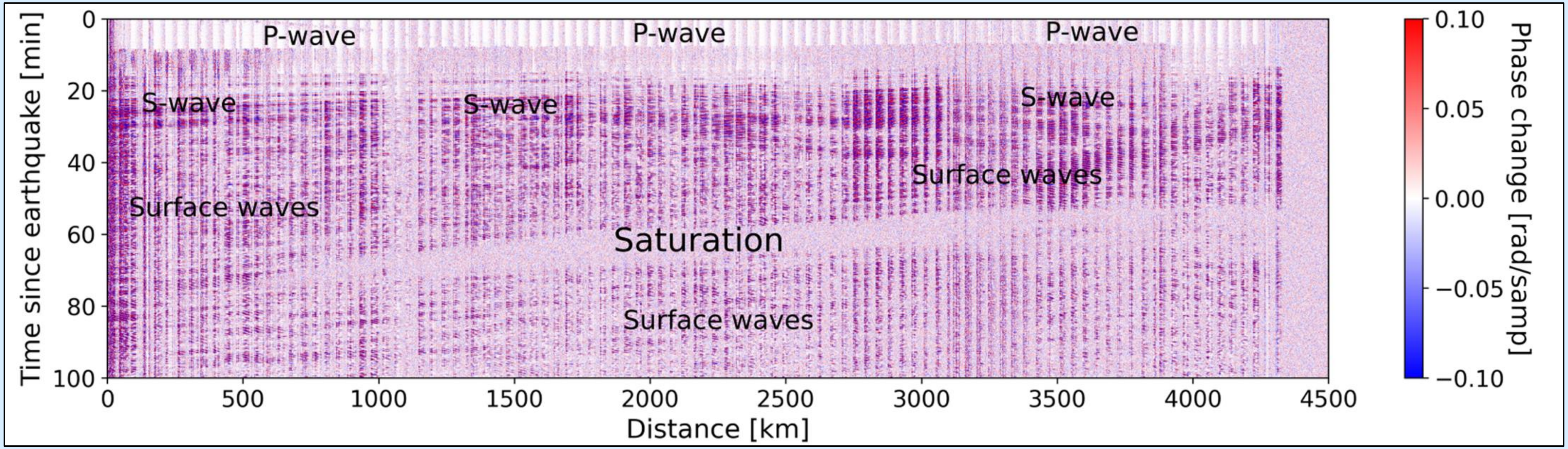






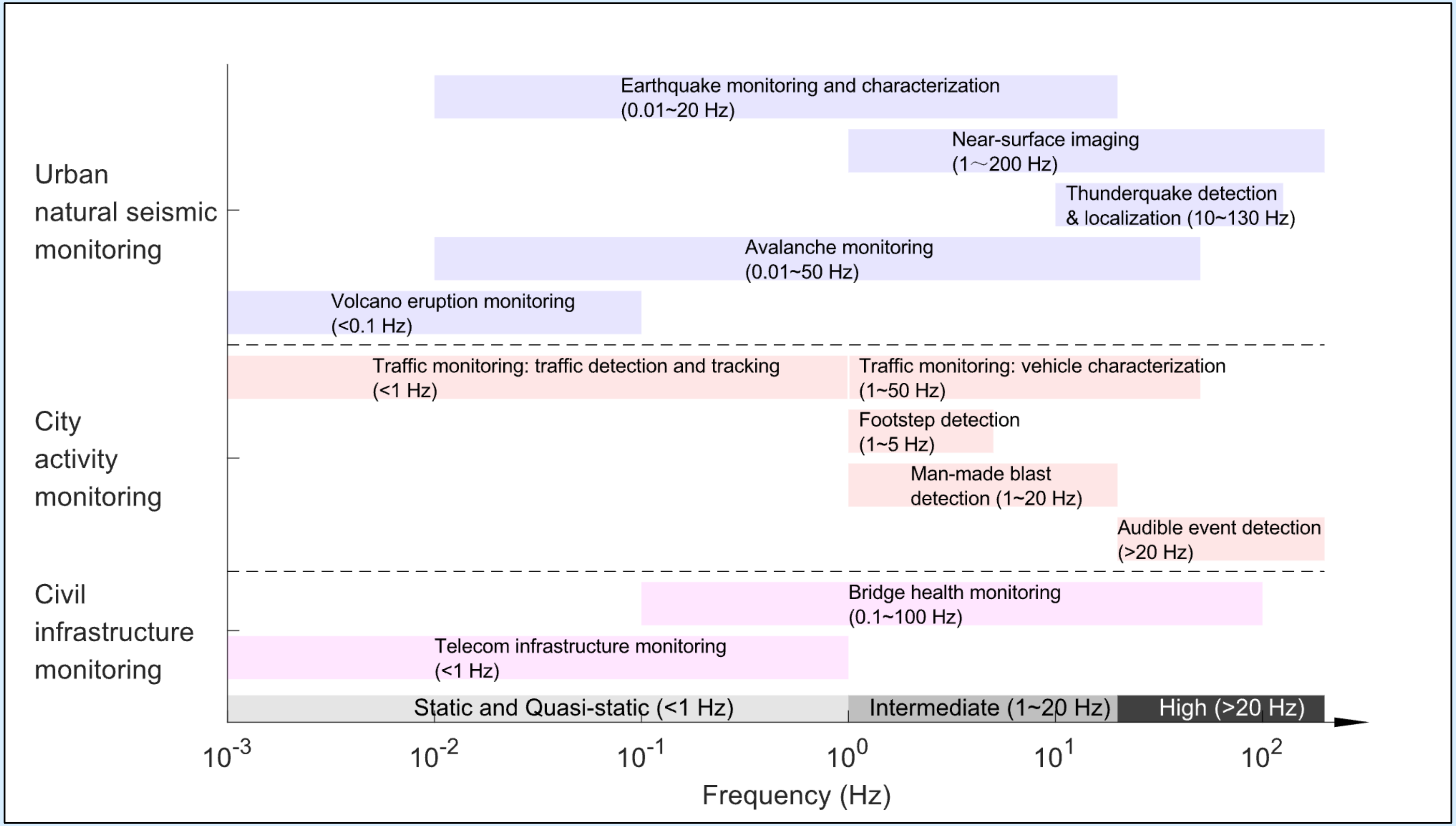
The future of fiber sensing





Mazur et al., 2025

DAS: a tool bridge scales



Liu et al., 2025

Acknowledgements

Google Cloud

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Thank you for your attention!
Questions?